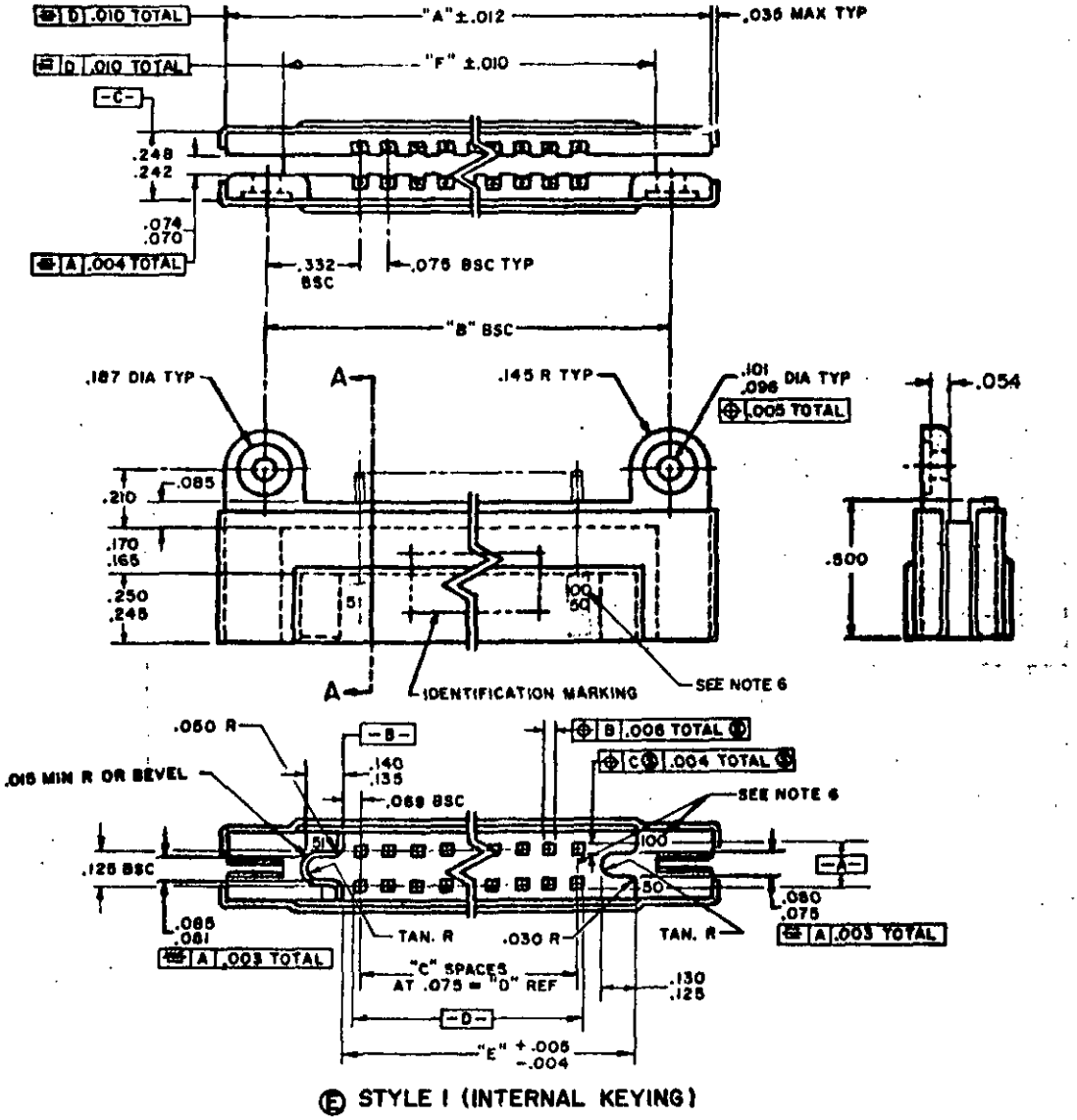
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, DECADE INCREMENT 10 THRU 120, 150, 160, AND 180 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.075 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-55302.



Ⓔ STYLE I (INTERNAL KEYING)

FIGURE 1. Connectors, plug, (.075 spacing).

Ⓔ denotes changes

FSC 5935

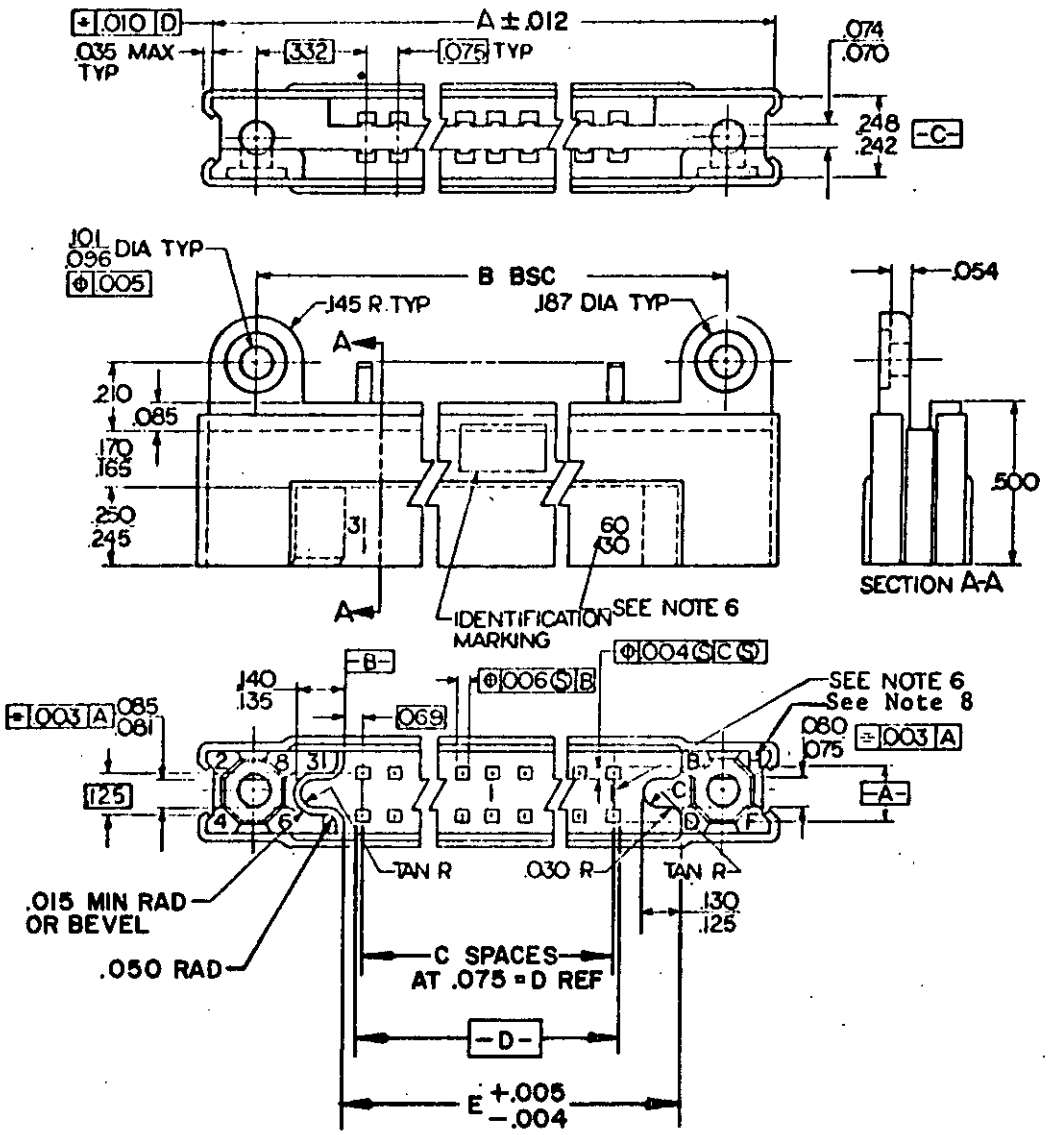
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1/ Dash number	No. of contacts	Dimensions						Contact identification numbers
		A	B	C	D	E	F	
01	10	1.255 (31.88)	.965 (24.51)	4	.300 (7.62)	.575 (14.61)	.855 (21.72)	1 - 5 6 - 10
02	20	1.630 (41.40)	1.340 (34.04)	9	.675 (17.15)	.950 (24.13)	1.230 (31.24)	1 - 10 11 - 20
03	30	2.005 (50.93)	1.715 (43.56)	14	1.050 (26.67)	1.325 (33.66)	1.605 (40.77)	1 - 15 16 - 30
04	40	2.380 (60.45)	2.090 (53.09)	19	1.425 (36.20)	1.700 (43.18)	1.980 (50.29)	1 - 20 21 - 40
05	50	2.755 (69.98)	2.465 (62.61)	24	1.800 (45.72)	2.075 (52.71)	2.355 (59.82)	1 - 25 26 - 50
06	60	3.130 (79.50)	2.840 (72.14)	29	2.175 (55.25)	2.450 (62.23)	2.730 (69.34)	1 - 30 31 - 60
07	70	3.505 (89.03)	3.215 (81.66)	34	2.550 (64.77)	2.825 (71.76)	3.105 (78.87)	1 - 35 36 - 70
08	80	3.880 (98.55)	3.590 (91.19)	39	2.925 (74.30)	3.200 (81.28)	3.480 (88.39)	1 - 40 41 - 80
09	90	4.255 (108.08)	3.965 (100.71)	44	3.300 (83.82)	3.575 (90.81)	3.855 (97.92)	1 - 45 46 - 90
10	100	4.630 (117.60)	4.340 (110.24)	49	3.675 (93.35)	3.950 (100.33)	4.230 (107.44)	1 - 50 51 - 100
11	110	5.005 (127.13)	4.715 (119.76)	54	4.050 (102.87)	4.325 (109.86)	4.605 (116.97)	1 - 55 56 - 110
12	120	5.380 (136.65)	5.090 (129.29)	59	4.425 (112.40)	4.700 (119.38)	4.980 (126.49)	1 - 60 61 - 120
13	150	6.505 (165.23)	6.215 (157.86)	74	5.550 (140.97)	5.825 (147.96)	6.105 (155.07)	1 - 75 76 - 150
14	160	6.880 (174.75)	6.590 (167.39)	79	5.925 (150.50)	6.200 (157.48)	6.480 (164.59)	1 - 80 81 - 160
15	180	7.630 (193.80)	7.340 (186.44)	89	6.675 (169.55)	6.950 (176.53)	7.230 (183.64)	1 - 90 91 - 180

(E) 1/ Dash numbers 01 thru 15 are for style 1 (internal keying) connectors.

FIGURE 1. Connectors, plug, (.075 spacing) - Continued.

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STYLE 2 (EXTERNAL KEYING)

Ⓔ FIGURE 1. Connectors, plug, (.075 spacing).

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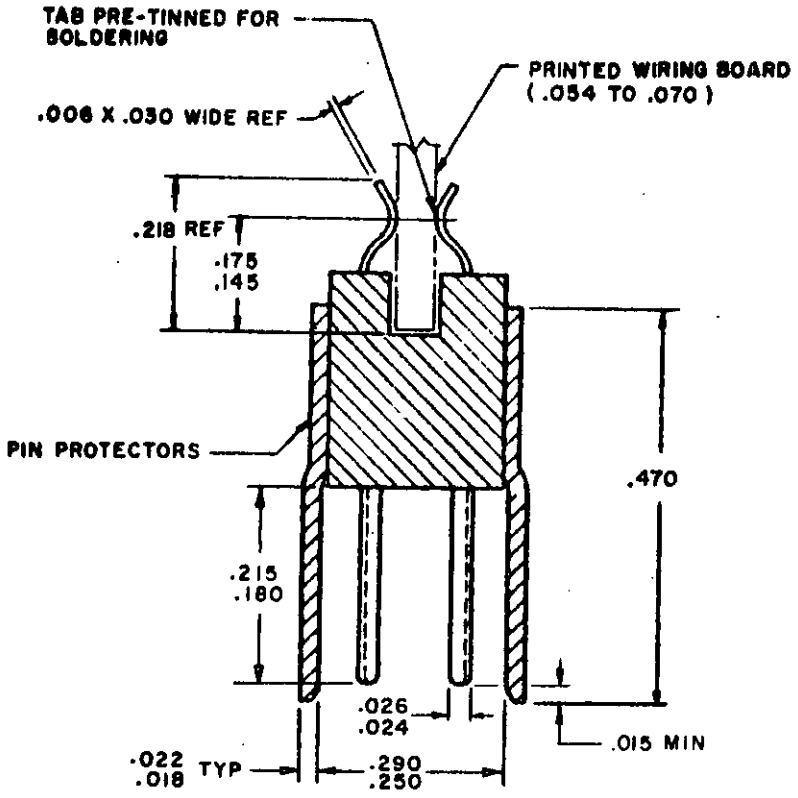
Ⓔ

1/ Dash number	No. of contacts	Dimensions					Contact identification numbers
		A	B	C	D	E	
16	10	1.255 (31.88)	.965 (24.51)	4	.300 (7.62)	.575 (14.61)	1 - 5 6 - 10
17	20	1.630 (41.40)	1.340 (34.04)	9	.675 (17.15)	.950 (24.13)	1 - 10 11 - 20
18	30	2.005 (50.93)	1.715 (43.56)	14	1.050 (26.67)	1.325 (33.66)	1 - 15 16 - 30
19	40	2.380 (60.45)	2.090 (53.09)	19	1.425 (36.20)	1.700 (43.18)	1 - 20 21 - 40
20	50	2.755 (69.98)	2.465 (62.61)	24	1.800 (45.72)	2.075 (52.71)	1 - 25 26 - 50
21	60	3.130 (79.50)	2.840 (72.14)	29	2.175 (55.25)	2.450 (62.23)	1 - 30 31 - 60
22	70	3.505 (89.03)	3.215 (81.66)	34	2.550 (64.77)	2.825 (71.76)	1 - 35 36 - 70
23	80	3.880 (98.55)	3.590 (91.19)	39	2.925 (74.30)	3.200 (81.28)	1 - 40 41 - 80
24	90	4.255 (108.08)	3.965 (100.71)	44	3.300 (83.82)	3.575 (90.81)	1 - 45 46 - 90
25	100	4.630 (117.60)	4.340 (110.24)	49	3.675 (93.35)	3.950 (100.33)	1 - 50 51 - 100
26	110	5.005 (127.13)	4.715 (119.76)	54	4.050 (102.87)	4.325 (109.86)	1 - 55 56 - 110
27	120	5.380 (136.65)	5.090 (129.29)	59	4.425 (112.40)	4.700 (119.38)	1 - 60 61 - 120
28	150	6.505 (165.23)	6.215 (157.86)	74	5.550 (140.97)	5.825 (147.96)	1 - 75 76 - 150
29	160	6.880 (174.75)	6.590 (167.39)	79	5.925 (150.50)	6.200 (157.48)	1 - 80 81 - 160
30	180	7.630 (193.80)	7.340 (186.44)	89	6.675 (169.55)	6.950 (176.53)	1 - 90 91 - 180

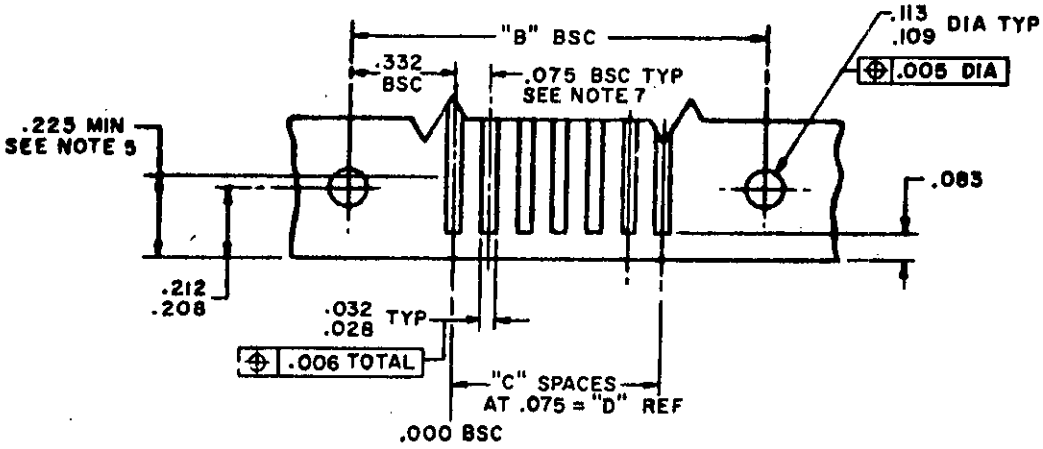
Ⓔ 1/ Dash numbers 16 thru 30 are for style 2 (external keying) connectors.

Ⓔ FIGURE 1. Connectors, plug, (.075 spacing) - Continued.

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SECTION A-A



RECOMMENDED PRINT WIRING BOARD FOR STYLES 1 AND 2

FIGURE 1. Connectors, plug, (.075 spacing) - Continued.

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INCHES	MM	INCHES	MM
.003	.08	.101	2.57
.004	.10	.109	2.77
.005	.13	.113	2.87
.006	.15	.125	3.18
Ⓔ .008	.20	.130	3.30
.010	.25	.135	3.43
.012	.30	.140	3.56
.015	.38	.145	3.68
.018	.46	.165	4.19
.022	.56	.170	4.32
.024	.61	.175	4.44
.026	.66	.180	4.57
.028	.71	.187	4.75
.030	.76	.208	5.28
.032	.81	.210	5.33
.035	.89	.212	5.38
.050	1.27	.215	5.46
.054	1.37	.218	5.54
.069	1.75	.225	5.72
.070	1.78	.242	6.15
.074	1.88	.245	6.22
.075	1.90	.248	6.30
.080	2.03	.250	6.35
.081	2.06	.290	7.37
.083	2.11	.332	8.43
.085	2.16	.470	11.94
.096	2.44	.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/24 or MIL-C-55302/25.
5. Minimum pad length on printed circuit board.
6. Numbers indicating every five cavities stamped on side (except 10 position). Numbers indicating end cavities and lines indicating every five cavities stamped on mating face.
7. Termination layout on .025 (.64 mm) modular grid.
Ⓔ 8. Location indicators embossed on surface.

FIGURE 1. Connectors, plug, (.075 spacing) - Continued.

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REQUIREMENTS:

(E) Design and construction:

Dimensions and configuration: See figure 1.

Material:
Plating: The contact plating for the engagement area, .230 inches minimum length, shall be gold per MIL-G-45204, type II, grade C, class 1 over nickel per QQ-N-290, class 2, 30 to 150 microinches. The contact plating for the solder tail area, .160 inches minimum length, shall be tin lead per MIL-T-10727, type I, 100 microinches minimum over nickel per QQ-N-290, class 2, 30 to 150 microinches with a tin lead coat per MIL-T-10727, type II, composition SN60 or SN63 of QQ-S-571 for .080 inches minimum length. The termination end of the contact must meet method 208 of MIL-STD-202 with the exception that the solder dip temperature shall be 500° ±10°F.

Contact identification: See figure 1.

Coating: The aluminum pin protector shall be coated per MIL-A-8625, type 1 or epoxy coated .001 to .003 inches thick.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per connector at room ambient.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:
Sea level: 900 volts rms.
High altitude: 200 volts rms.

Part number: M55302/23- (dash number from figure 1).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Navy - EC
Air Force - 17

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Navy - SH, AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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